Online Appendix

for

The Evolution of Market Power in the Automobile Industry

by

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A Demand Estimation Procedure

In this appendix, we detail the three steps of our demand estimation procedure.

A.1 Unobserved Heterogeneity and Mean Valuation

Following Berry et al. (1995) we can decompose a consumer utility net of taste for the outside good into a vertical component δ_{jt} and horizontal components,[23]

$$u_{ijt} - \gamma_t = \delta_{jt} + \mu_{ijt}(\theta) + \epsilon_{ijt}.$$

Where the vertical component is

$$\delta_{jt} = \bar{\beta}\mathbf{x}_{jt} + \bar{\alpha}p_{jt} + \xi_{jt} - \gamma_t \tag{9}$$

and the heterogeneity term is

$$\mu_{ijt}(\theta) = \sum_k \sum_h \beta_{kh} x_{jt}^k D_{it}^h + \sum_h \alpha_h p_j D_{it}^h + \sum_k \sigma_k x_{jt}^k \nu_{ik}, \tag{10}$$

where x_{jt}^k is the kth element of $\mathbf{x}_{jt}$ and we collect the heterogeneity parameters into the vector $\theta = (\{\beta_{kh}\}, \{\alpha_h\}, \sigma)$.

Our goal in this step is to estimate (θ, δ). For any consumer i, the conditional choice probability as a function of parameters is

$$\mathcal{s}_{ijt}(\theta, \delta) = \frac{\exp\left(\delta_{jt} + \mu_{ijt}(\theta)\right)}{1 + \sum_{k \in J_t} \exp\left(\delta_{kt} + \mu_{ikt}(\theta)\right)}. \tag{11}$$

Integrating these choice probabilities over the distribution of consumers gives us the market shares. Since there is a one-to-one mapping between δ and market shares, we can solve for mean valuations as a function of θ by matching model predicted shares to the market share data,

$$s_{jt} = \int \mathcal{s}_{ij}(\theta, \delta(\theta)) dF_t(i).$$

[23] In this stage of estimation, it is convenient to re-normalize utility to be net of the outside good in year t, so that γ_t is a term in δ_{jt}. We will show how to estimate γ_t below.

We can now construct the moments for our estimator of θ. Let $\mathcal{s}_{ij}(\theta) = \mathcal{s}_{ij}(\theta, \delta(\theta))$. For readability, we drop t from the notation from the rest of this section and let y_i be the observed purchase of consumer i

Our first set of moments rely on microdata where we observe consumers' automobile choice as well as their demographic characteristics, so we observe a random sample $\{y_i, \mathbf{D}_i\}$. We use this information to match product characteristics conditional on consumer demographics. Specifically, we construct moments of the form[24]

$$\mathbf{g}_1(\theta) = E[\widehat{\mathbf{x}_{(y_i)}|i \in \mathscr{H}}] - \int \sum_j \mathbf{x}_j \mathcal{s}_{ij}(\theta)\ dF(i|i \in \mathscr{H}), \tag{12}$$

where $\mathscr{H}$ describes a set of consumers identifiable based on demographics, $\mathbf{x}_{(y_i}$ is the characteristic vector of the product purchased by i and $E[\widehat{\mathbf{x}_{(y_i)}|i \in \mathscr{H}}]$ is an estimate from the microdata. In practice, we match differences and ratios of $E[\widehat{\mathbf{x}_{(y_i)}|i \in \mathscr{H}}]$ across alternative demographic sets, $\mathscr{H}$. Table A.4 lists the demographic moments we target and the associated model fit.

Our second set of moments relies on microdata for which we observe the consumers first and second choices of products. That is, the data is a random sample $\{y_i, z_i\}$, where z_i is the stated second choice of consumer i. Conditional on purchasing an automobile, our model predicts the first and second choices of consumer i,[25]

$$\mathcal{s}_{i(j,k)}(\theta) = \frac{\exp\left(\delta_j(\theta) + \mu_{ij}(\theta)\right)}{\sum_{\ell \in J} \exp\left(\delta_\ell + \mu_{i\ell}(\theta)\right)} \cdot \frac{\exp\left(\delta_k(\theta) + \mu_{ik}(\theta)\right)}{\sum_{\ell \in J \setminus j} \exp\left(\delta_\ell + \mu_{i\ell}(\theta)\right)}. \tag{13}$$

We construct moments based on the correlation of product characteristics of first and second choices,

$$\mathbf{g}_2(\theta) = E[\widehat{\mathbf{x}_{(y_i)} \circ \mathbf{x}_{(z_i)}}] - \int \sum_{j,k} (\mathbf{x}_j \circ \mathbf{x}_k) \mathcal{s}_{i(j,k)}(\theta) dF(i) \tag{14}$$

where $\circ$ denotes element-wise multiplication and $E[\widehat{\mathbf{x}_{(y_i)} \circ \mathbf{x}_{(z_i)}}]$ is an estimate based on the microdata. Table VI displays the second choice correlations we target and the model fit.

We stack these two sets of moments and estimate θ via simulated GMM. We use a weight matrix based on the inverse variance matrix of the data moments. Simulation over the distribution of consumers follows Pakes and Pollard (1989). Given $\hat{\theta}$, our estimate of mean valuations is $\hat{\delta} = \delta(\hat{\theta})$.

[24] In practice, we condition this moment on purchasing an automobile, since the outside good does not have characteristics. An exception to this is that we do include one moment based on purchase probabilities in order to estimate a demographic coefficient on the constant.

[25] Our second choice data does not include information on outside good selection, so we again condition out the no purchase option when constructing second choice moments.

A.2 Mean Taste for Characteristics

With the estimates of mean valuations from the previous step, we can now estimate mean tastes for product characteristics. We use the following regression equation,

$$\delta_{jt} = \bar{\beta}\mathbf{x}_{jt} + \bar{\alpha}p_{jt} + \iota_t + \tilde{\xi}_{jt}, \tag{15}$$

where $\iota_t = \tau_t - \gamma_t$ absorbs the effect of the average utility of the outside good and the average car quality in year t. We use our first stage estimate $\hat{\delta}$ as a proxy for δ and employ a simple (IV) regression where the real exchange rate is our instrument for price. The extension to allow for variation in mean preference parameters over time is straightforward.

A.3 Mean Quality over Time

Our final step estimates τ and γ separately using the continuing product condition (2). The empirical analogue of this condition can be rewritten as an estimator of τ_t using the residuals from our second step,

$$\hat{\tau}_t = \hat{\tau}_{t-1} + \sum_{j \in \mathscr{C}_t} (\hat{\tilde{\xi}}_{jt-1} - \hat{\tilde{\xi}}_{jt}), \tag{16}$$

with τ_0 normalized to 0. Finally, we can estimate $\hat{\gamma}_t = \hat{\iota}_t - \hat{\tau}_t$.

B Robustness Specifications

In this subsection, we describe various alternative utility specifications which lead to the alternative markup estimates that appear in Figure VIb.

Our baseline estimates, only employ RXR as an instrument for price to be as transparent as possible regarding the source of variation identifying price sensitivity. However, for the robustness specifications that allow for more flexible preferences over time, we require additional instruments. For these additional instruments, we use a dummy for country of assembly. Since we control for brand as part of utility, this instrument is similar to Gandhi and Houde (2020) who use a dummy for when a vehicle's production is moved to the U.S., the difference being we also capture productivity differences across non-US locations.[26] The theoretical justification for this instrument is similar to the RXR instrument in that vehicles' production likely reflects cost advantages or disadvantages of that location which may not be captured in the RXR. For example, a foreign manufacturer might relocate production to Canada where labor costs are higher than in Mexico, but transportation costs into major markets in the U.S. are lower. These components of cost should be reflected in price.

[26] Adding these instruments to the baseline specification leads to almost identical estimates of elasticities and markups.

B.1 Over-time flexibility in price sensitivity.

We estimate alternative specifications that allow the mean price sensitivity parameter, $\bar{\alpha}$, to vary over time. Price parameter results are reported in Table A.1, and vehicle attribute parameters are reported in Table A.2. Compared to our baseline, Column (1), we allow for a linear time trend on $\bar{\alpha}$ in the specification in Column (2). In Column (3) we allow $\bar{\alpha}$ to vary every five years. In both of these specifications, we also allow the coefficients on the vehicle style dummies to have a trend. However, all other parameters are fixed over time.

Table A.1: "Mean" price parameters for different specifications

	(1) Baseline	(2) Linear Trend	(3) 5-year Breaks	(4) Logged Price	(5) Log, 5yr Breaks
Price	-3.112	-2.839	–	-9.359	–
	(1.124)	(0.457)		(3.850)	
Trend X Price	–	-0.018	–	–	–
		(0.011)			
1980-1984	–	–	-1.933	–	-6.283
			(0.472)		(1.529)
1985-1989	–	–	-3.050	–	-11.541
			(0.425)		(1.573)
1990-1994	–	–	-2.682	–	-10.244
			(0.364)		(1.603)
1995-1999	–	–	-2.722	–	-10.707
			(0.381)		(1.784)
2000-2004	–	–	-2.570	–	-10.136
			(0.459)		(1.897)
2005-2009	–	–	-3.222	–	-12.127
			(0.493)		(2.166)
2010-2014	–	–	-3.650	–	-15.011
			(0.536)		(2.328)
2015-2018	–	–	-3.312	–	-12.857
			(0.526)		(2.293)
Add assembly IVs		✓	✓		✓

NOTE: Brand and year dummies included. Standard errors account for correlation within make of realizations of ξ_{jt} and are bootstrapped over the entire estimation procedure, so they account for sampling error in the survey data. Price is in 2015 $10,000.

B.2 Log of price in indirect utility

We also estimate a version of the model where price enters logged into consumer utility. The alternative utility specification is therefore

$$u_{ijt} = \beta_{it}\mathbf{x}_{jt} + \alpha_{it}\log(p_{jt}) + \xi_{jt} + \epsilon_{ijt}. \tag{17}$$

We estimate one version of this specification with a time-invariant $\bar{\alpha}$ (Column 4), and another version where we allow $\bar{\alpha}$ to vary every five years (Column 5). The specification with logged price has different coefficient estimates for the observed and unobserved heterogeneity because price enters the "non-linear" part of the utility. Those coefficient estimates are reported in Table A.3. Overall the sizes and magnitudes of the parameters are similar to the baseline specification for when price is not involved. The one exception is that the interaction between income and the constant is not precisely estimated.

B.3 Three separate time periods

Lastly, we estimate the entire model separately for different time periods. Since our micro data does not have complete coverage over our entire sample, we are somewhat limited with how flexible we can allow all of the parameters to be. Our second choice data are for 1991, 1999, 2005, and 2015. Additionally, our consumer survey data have different coverage. We split our sample into three nearly equal time periods and estimate the entire model separately for each segment: 1980-1992, 1993-2004, and 2005-2018. Within each time segment, we restrict all parameters to be fixed. As with all our estimates that allow price sensitivity to vary over time, we include both RXR and assembly location dummies as instruments. The share-weighted average markups associated with this specification is the light dotted line in Figure VI. The parameter estimates for this specification are available from the authors.

C Additional Results and Analysis

C.1 Model Fit

In this section, we report model fit for all of the moments used in estimation. Fit for the demographic interaction moments, for both the MRI and CEX samples, are reported in Table A.4. Table A.5 reports the second choice moments we match in demand estimation. An aggregated (averaged across the four sample years) version of these moments appears in the body of the paper (Table VI).

We also display model fit for the alternative model specification described in Appendix B, where we define indirect utility in terms of $\log(price)$. The corresponding model fit tables are Tables A.6 and Table A.7.

C.2 Determinants of Marginal Cost

In a subset of the counterfactual exercises, we consider scenarios that alter the marginal cost of products. To do so, we estimate a parsimonious model of the determinants of marginal cost, relating them to observable characteristics, the real exchange rate, and a linear time trend to capture technological innovation.

Table A.2: "Mean" attribute parameters for different specifications

	(1) Baseline	(2) Linear Trend	(3) 5-year Breaks	(4) Logged Price	(5) Log, 5yr Breaks
Height	-1.788	-1.696	-1.631	-1.916	-1.961
	(0.319)	(0.248)	(0.231)	(0.362)	(0.253)
Footprint	0.534	0.409	0.502	0.740	0.564
	(0.261)	(0.252)	(0.240)	(0.246)	(0.237)
Horsepower	1.018	1.240	1.189	0.633	1.421
	(0.954)	(0.491)	(0.413)	(0.882)	(0.420)
MPG	-0.965	-0.957	-0.971	-0.894	-0.766
	(0.211)	(0.210)	(0.203)	(0.192)	(0.180)
Curbweight	0.339	0.454	0.088	0.164	0.577
	(0.936)	(0.498)	(0.438)	(1.006)	(0.570)
Num. of Trims	1.118	1.108	1.153	1.251	1.222
	(0.141)	(0.081)	(0.072)	(0.105)	(0.066)
Release Year	-0.238	-0.296	-0.234	-0.144	-0.203
	(0.160)	(0.130)	(0.120)	(0.141)	(0.108)
Yrs. Since Design	-0.214	-0.220	-0.221	-0.230	-0.240
	(0.025)	(0.025)	(0.025)	(0.024)	(0.024)
Sport	-3.046	-3.039	-3.203	-3.464	-3.427
	(0.549)	(0.271)	(0.260)	(0.510)	(0.252)
Electric	-5.549	-5.244	-5.261	-5.433	-4.524
	(1.406)	(1.013)	(0.956)	(1.175)	(0.670)
Truck	-7.463	-6.885	-6.387	-9.566	-8.778
	(0.898)	(0.748)	(0.729)	(0.855)	(0.774)
SUV	-0.079	-2.727	-2.604	0.577	-2.620
	(0.339)	(0.582)	(0.565)	(0.257)	(0.554)
Van	-7.614	-5.942	-5.701	-5.775	-3.752
	(0.598)	(0.654)	(0.632)	(0.427)	(0.544)
Van Trend	–	-0.099	-0.106	–	-0.111
		(0.018)	(0.017)		(0.017)
SUV Trend	–	0.093	0.092	–	0.121
		(0.025)	(0.023)		(0.022)
Truck Trend	–	-0.042	-0.058	–	-0.050
		(0.020)	(0.019)		(0.017)
Add assembly IVs		✓	✓		✓

NOTE: Brand and year dummies included. Brand and year dummies included. Standard errors account for correlation within make of realizations of ξ_{jt} and are bootstrapped over the entire estimation procedure, so they account for sampling error in the survey data. Footprint is vehicle length times height in square inches. All continuous physical attributes are logged and standardized.

Table A.3: Coefficient Estimates, Logged Price

		Demographic Interactions						
	σ	Income	Inc. Sq.	Age	Rural	Fam. Size 2	FS 3-4	FS 5+
log(Price)	–	0.309	-1.784	6.280	–	–	–	–
		(0.015)	(0.081)	(0.293)				
Van	5.867	–	–	–	–	1.187	2.915	5.003
	(0.114)					(0.119)	(0.117)	(0.156)
SUV	4.205	–	–	–	–	–	–	–
	(0.073)							
Truck	9.265	–	–	–	3.935	–	–	–
	(0.394)				(0.409)			
Footprint	2.264	–	–	–	–	0.473	0.457	0.652
	(0.051)					(0.047)	(0.048)	(0.066)
Horsepower	1.337	–	–	–	–	–	–	–
	(0.092)							
Miles/Gal.	1.679	–	–	–	–	–	–	–
	(0.054)							
Luxury	2.827	–	–	–	–	–	–	–
	(0.026)							
Sport	2.787	–	–	–	–	–	–	–
	(0.053)							
EV	3.770	–	–	–	–	–	–	–
	(0.122)							
Euro. Brand	2.018	–	–	–	–	–	–	–
	(0.032)							
US Brand	2.320	–	–	–	–	–	–	–
	(0.027)							
Constant	–	-0.190	–	–	–	–	–	–
		(0.140)						

NOTE: Brand and year dummies included. Standard errors are constructed by bootstrapping the microdata. All continuous car characteristics are in logs and standardized, and price is in 2015 $10,000. Footprint is vehicle length times height in square inches. Income is normalized to have zero mean and unit variance.

Table A.4: Moments and Model Fit: Demographic Interactions

		MRI			CEX		
Demographic	Car Attribute	Data	Model	Only Demos	Data	Model	Only Demos
$\mathbb{E}[x\|Income\,Q_5] - \mathbb{E}[x\|Income\,Q_1]$	Price	0.215	0.383	0.363	0.603	0.400	0.421
$\mathbb{E}[x\|Income\,Q_4] - \mathbb{E}[x\|Income\,Q_1]$	Price	0.016	0.229	0.215	0.356	0.266	0.268
$\mathbb{E}[x\|Income\,Q_3] - \mathbb{E}[x\|Income\,Q_1]$	Price	-0.080	0.131	0.121	0.189	0.149	0.150
$\mathbb{E}[x\|Income\,Q_2] - \mathbb{E}[x\|Income\,Q_1]$	Price	-0.146	0.068	0.062	0.069	0.078	0.075
$\mathbb{E}[x\|Age > 60] - E[x\|Age < 30]$	Price	0.518	0.369	0.328	0.257	0.344	0.342
$\mathbb{E}[x\|Age50 - 60] - E[x\|Age < 30]$	Price	0.388	0.340	0.330	0.239	0.291	0.316
$\mathbb{E}[x\|Age40 - 50] - E[x\|Age < 30]$	Price	0.329	0.294	0.295	0.265	0.236	0.246
$\mathbb{E}[x\|Age30 - 40] - E[x\|Age < 30]$	Price	0.266	0.146	0.148	0.265	0.135	0.139
$\mathbb{E}[x\|Family = 2] - \mathbb{E}[x\|Family = 1]$	Van	0.025	0.022	0.022	0.020	0.021	0.021
$\mathbb{E}[x\|Family = 3/4] - \mathbb{E}[x\|Family = 1]$	Van	0.063	0.062	0.061	0.058	0.059	0.059
$\mathbb{E}[x\|Family = 5+] - \mathbb{E}[x\|Family = 1]$	Van	0.132	0.131	0.126	0.120	0.121	0.122
$\mathbb{E}[x\|Family = 2] - \mathbb{E}[x\|Family = 1]$	Footprint	0.036	0.027	0.026	0.026	0.027	0.027
$\mathbb{E}[x\|Family = 3/4] - \mathbb{E}[x\|Family = 1]$	Footprint	0.018	0.024	0.023	0.026	0.025	0.025
$\mathbb{E}[x\|Family = 5+] - \mathbb{E}[x\|Family = 1]$	Footprint	0.030	0.035	0.033	0.036	0.036	0.036
$\mathbb{E}[x\|Rural] - \mathbb{E}[x\|NotRural]$	Truck	-0.103	-0.098	-0.101	–	–	–
$\mathbb{E}[x\|Income\,Q_2] \,/\, \mathbb{E}[x\|Income\,Q_1]$	PurchaseProb	–	–	–	7.813	9.239	9.091
$\mathbb{E}[x\|Income\,Q_3] \,/\, \mathbb{E}[x\|Income\,Q_1]$	PurchaseProb	–	–	–	5.466	5.077	5.246
$\mathbb{E}[x\|Income\,Q_4] \,/\, \mathbb{E}[x\|Income\,Q_1]$	PurchaseProb	–	–	–	3.641	2.706	2.866
$\mathbb{E}[x\|Income\,Q_5] \,/\, \mathbb{E}[x\|Income\,Q_1]$	PurchaseProb	–	–	–	2.265	1.660	1.684

NOTE: Moments from the consumer samples that we target in estimation, along with the analog from our model at the estimated parameters. For the demographic moments, our data comes from two surveys, the Consumer Expenditure Survey (CEX) covering years 1980-2005 and MRI covering years 1992-2018. The "Only Demos" column correspond to a specification where we include all of the demographic interactions but none of the random coefficients.

Table A.5: Moments and Model Fit: Second Choice Moments

	1991		1999		2005		2015	
Attribute	Data	Model	Data	Model	Data	Model	Data	Model
Van	0.688	0.734	0.735	0.739	0.735	0.719	0.720	0.696
SUV	0.605	0.580	0.652	0.639	0.623	0.665	0.690	0.663
Truck	0.840	0.792	0.830	0.790	0.828	0.805	0.872	0.799
Footprint	0.638	0.665	0.680	0.684	0.739	0.713	0.782	0.714
Horsepower	0.512	0.542	0.567	0.559	0.643	0.611	0.674	0.636
MPG	0.627	0.647	0.686	0.667	0.664	0.664	0.611	0.641
Luxury	0.426	0.514	0.438	0.469	0.521	0.499	0.550	0.476
Sport	0.229	0.266	0.217	0.244	0.314	0.267	0.348	0.338
Electric	–	–	–	0.003	–	–	0.373	0.371
Euro Brand	0.186	0.276	0.337	0.359	0.409	0.368	0.413	0.363
US Brand	0.468	0.453	0.475	0.471	0.510	0.474	0.464	0.480

NOTE: Moments from the Maritz second choice survey that we target in estimation and the analog from our model at the estimated parameters.

$$\log(c_{jt}) = \mu \mathbf{x}_{jt} + \psi \cdot \text{RXR}_{jt} + \rho \cdot t + \omega_{jt}, \tag{18}$$

Where $\mathbf{x}_{jt}$ is the characteristic set used in utility and RXR_{jt} is the real exchange rate (our cost instrument).[27] The coefficient estimates for this estimation are provided in Table A.8.

For the exercises in counterfactual Mechanism 4, which are reported in Figure XVII, we compute counterfactual marginal costs by eliminating the technological improvements represented by the

[27] We have also experimented with including unobserved quality ξ_{jt} in the specification for marginal cost. While this produces qualitatively similar results for the counterfactuals, we opted to omit ξ from our baseline specification since it incorporates a mix of technological improvements many of which (such as inventing anti-lock breaks in 1980) are likely not directly attributable to marginal costs.

Table A.6: Moments and Model Fit: Demographic Interactions for log($price$) Specification

		MRI		CEX	
Demographic	Car Attribute	Data	Model	Data	Model
$\mathbb{E}[x\|Income\,Q_5] - \mathbb{E}[x\|Income\,Q_1]$	Price	0.215	0.121	0.603	0.123
$\mathbb{E}[x\|Income\,Q_4] - \mathbb{E}[x\|Income\,Q_1]$	Price	0.016	0.088	0.356	0.098
$\mathbb{E}[x\|Income\,Q_3] - \mathbb{E}[x\|Income\,Q_1]$	Price	-0.08	0.055	0.189	0.061
$\mathbb{E}[x\|Income\,Q_2] - \mathbb{E}[x\|Income\,Q_1]$	Price	-0.146	0.029	0.069	0.033
$\mathbb{E}[x\|Age > 60] - E[x\|Age < 30]$	Price	0.518	0.12	0.257	0.108
$\mathbb{E}[x\|Age50 - 60] - E[x\|Age < 30]$	Price	0.388	0.119	0.239	0.1
$\mathbb{E}[x\|Age40 - 50] - E[x\|Age < 30]$	Price	0.329	0.113	0.265	0.085
$\mathbb{E}[x\|Age30 - 40] - E[x\|Age < 30]$	Price	0.266	0.059	0.265	0.051
$\mathbb{E}[x\|Family = 2] - \mathbb{E}[x\|Family = 1]$	Van	0.025	0.022	0.02	0.021
$\mathbb{E}[x\|Family = 3/4] - \mathbb{E}[x\|Family = 1]$	Van	0.063	0.061	0.058	0.059
$\mathbb{E}[x\|Family = 5+] - \mathbb{E}[x\|Family = 1]$	Van	0.132	0.129	0.12	0.121
$\mathbb{E}[x\|Family = 2] - \mathbb{E}[x\|Family = 1]$	Footprint	0.036	0.027	0.026	0.027
$\mathbb{E}[x\|Family = 3/4] - \mathbb{E}[x\|Family = 1]$	Footprint	0.018	0.023	0.026	0.024
$\mathbb{E}[x\|Family = 5+] - \mathbb{E}[x\|Family = 1]$	Footprint	0.03	0.034	0.036	0.036
$\mathbb{E}[x\|Rural] - \mathbb{E}[x\|NotRural]$	Truck	-0.103	-0.101	–	–
$\mathbb{E}[x\|Income\,Q_2] \,/\, \mathbb{E}[x\|Income\,Q_1]$	PurchaseProb	–	–	7.813	8.081
$\mathbb{E}[x\|Income\,Q_3] \,/\, \mathbb{E}[x\|Income\,Q_1]$	PurchaseProb	–	–	5.466	5.825
$\mathbb{E}[x\|Income\,Q_4] \,/\, \mathbb{E}[x\|Income\,Q_1]$	PurchaseProb	–	–	3.641	3.124
$\mathbb{E}[x\|Income\,Q_5] \,/\, \mathbb{E}[x\|Income\,Q_1]$	PurchaseProb	–	–	2.265	2.145

NOTE: Moments from the consumer samples that we target in estimation, along with the analog from our model at the estimated parameters. For the demographic moments, our data comes from two surveys, the Consumer Expenditure Survey (CEX) covering years 1980-2005 and MRI covering years 1992-2018. The "Only Demos" column correspond to a specification where we include all of the demographic interactions but none of the random coefficients.

Table A.7: Moments and Model Fit: Second Choice Moments, log($price$) Specification

	1991		1999		2005		2015	
Attribute	Data	Model	Data	Model	Data	Model	Data	Model
Van	0.688	0.735	0.735	0.738	0.735	0.720	0.720	0.679
SUV	0.605	0.567	0.652	0.655	0.623	0.677	0.690	0.681
Truck	0.840	0.834	0.830	0.836	0.828	0.840	0.872	0.835
Footprint	0.638	0.675	0.680	0.707	0.739	0.733	0.782	0.738
Horsepower	0.512	0.542	0.567	0.561	0.643	0.610	0.674	0.637
MPG	0.627	0.639	0.686	0.664	0.664	0.663	0.611	0.635
Luxury	0.426	0.512	0.438	0.467	0.521	0.493	0.550	0.481
Sport	0.229	0.252	0.217	0.253	0.314	0.279	0.348	0.338
Electric	–	–	–	0.004	–	–	0.373	0.371
Euro Brand	0.186	0.283	0.337	0.356	0.409	0.355	0.413	0.364
US Brand	0.468	0.451	0.475	0.476	0.510	0.482	0.464	0.490

NOTE: Moments from the Maritz second choice survey that we target in estimation and the analog from our model at the estimated parameters.

time trend (i.e., setting $\rho = 0$) while holding all other cost elements fixed (including the residuals). We then solve for a new price equilibrium and compute the resulting consumer surplus.

C.3 Additional Markup Counterfactuals

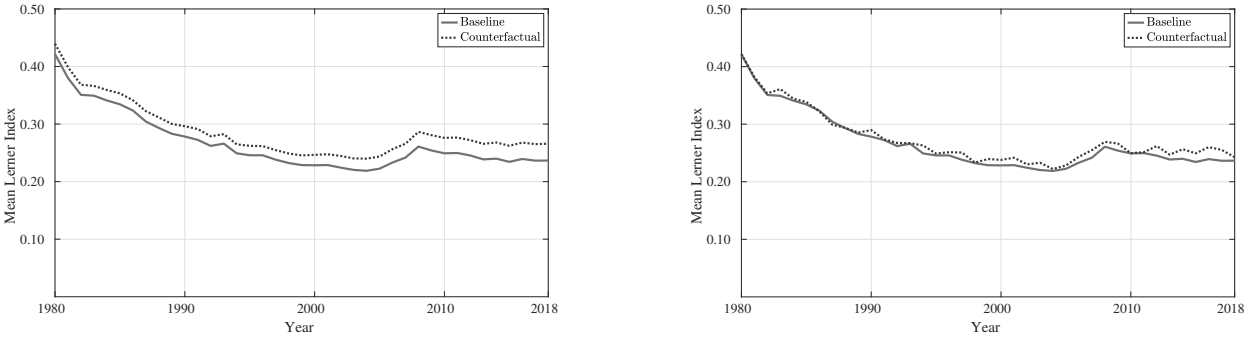
In this section we report plots for additional markup counterfactuals mentioned in Section 5.4.1. In Figure XVIIIa, we display median markups after assigning all foreign brands to be owned by one of the Big 3 U.S. manufacturers. Markups are slightly higher throughout the time period, but the downward trend remains. In Figure XVIIIb, we display median markups after (randomly) limiting the number of products offered each year to match the number of offerings in 1980. Median

Table A.8: Marginal Cost Function Estimates

	Coefficient	S.E.
Trend	-0.014	(0.002)
RXR	0.211	(0.030)
Height	-0.839	(0.133)
Footprint	-0.289	(0.122)
HP	0.552	(0.049)
MPG	0.096	(0.042)
Curbweight	1.280	(0.125)
# Trims	-0.032	(0.007)
Release Year	-0.012	(0.013)
SUV	0.025	(0.026)
Truck	-0.124	(0.037)
Van	0.042	(0.029)
EV	0.313	(0.066)
Sport	0.110	(0.019)
Design Years	-0.000	(0.003)
Brand Effect	✓	

NOTE: Dependent variable is log marginal costs recovered from the first-order conditions. All continuous attributes are logged. Standard errors are clustered by brand.

markups have a similar trend in this case as the the baseline.



(a) Markup Counterfactual: Foreign Ownership

(b) Markup Counterfactual: 1980 Product Count

Figure XVIII: Additional Counterfactual Markups

NOTE: Panel (a) displays counterfactual share weighted mean markups for the counterfactual described in Mechanism 1 in Section 6.2. Panel (b) displays median markups for the counterfactual described in Mechanism 2 in Section 6.2.

C.4 Used Car Analysis

We manually collected data from the Kelly Blue Book website in December 2021 for mint condition, the best possible option, which KBB reflects 1-2% of the vehicles they evaluate, used automobiles with a total of 500 miles driven produced every five years between 1992 and 2017. Our query asked for the private party transaction value of the top five vehicles in each of the years for each of

cars, SUV's, vans, and trucks. We then regressed the midpoint of Kelly Blue Book private party transaction value range against characteristics and dummy variables for the year of production. The year of production dummy variables should capture the unobserved product differences across years of production. Table A.9 reports the results, which indicate a steady increase in product quality over time which is consistent with the results of our demand estimation.

Table A.9: KBB Price Regression

	Coefficient	S.E.
Height	26058.27	(5244.96)
Footprint	12470.29	(6862.72)
Horsepower	600.47	(2407.91)
MPG	-258.53	(4450.29)
Weight	-9710.17	(5279.84)
No. Trims	-2.65	(11.90)
Years Since Redesign	-99.44	(100.48)
Truck	-1809.78	(1403.22)
SUV	421.12	(1223.43)
Van	-1124.60	(1101.78)
Year 1997	-586.04	(951.43)
Year 2002	395.62	(1251.81)
Year 2007	3424.69	(1506.16)
Year 2012	10224.18	(1485.64)
Year 2017	19638.88	(1956.91)

NOTE: Unit of observations: year make-model, from 1980 to 2018. Number of observations: 72. Specification include make fixed effects. Car characteristics in logs.